

GASPARD ABEL

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CURRENT POSITION

Ecole Normale Supérieure de Paris-Saclay, Gif-sur-Yvette

Sept. 2023 - Now

PhD student in Mathematics at Interfaces at the Centre Borelli.

Supervision: Julien Randon-Furling, Jean-Pierre Nadal, Argyris Kalogeratos

Hadamard Mathematics Doctoral School (EDMH)

Structural and semantic analysis of information flows in social networks

networks: modeling its underlying mechanisms and its influence on public opinion

ACADEMIC BACKGROUND

Ecole Normale Supérieure de Paris-Saclay, Gif-sur-Yvette

Sept. 2019 - Sep. 2023

Physics of complex Systems M2, Sorbonne, Université Paris-Cité,

Université Paris-Saclay. Courses included: advanced statistical physics,

stochastic processes, out-of-equilibrium physics, machine learning...

François-René de Chateaubriand high school, Rennes

Sept. 2017 - Sept. 2019

Intensive program preparing for the National competitive exams

for entry to top schools in Mathematics and Physics

SKILLS

Computer Science

Python, LaTeX, Matlab

Software & Tools

OpenMole (ABM exploration software), git

Language

English : C2

Spanish : B1

Italian : A2

SCIENTIFIC PRODUCTION

On the effect of asynchronous updates in social dilemmas,

Aug. 2026

G. Abel, I. Conjeaud and A. Kalogeratos working paper,

working paper

Algorithmic collusion under asynchronous price updating,

Aug. 2026

I. Conjeaud, G. Abel, A. Kalogeratos,

to be submitted

Dual-Criterion Curriculum Learning: Application to Temporal Data,

Mar. 2026

G. Abel, E. Campagne, M. Benloughmari, A. Kalogeratos.

to be submitted

Uncovering social network activity using joint user and topic interaction,

Jan. 2026

G. Abel, A. Kalogeratos, J.-P. Nadal, J. Randon-Furling

Published in *IEEE Transactions on Computational Social Systems*

Beyond ReLU: Bifurcation, Oversmoothing, and Topological Priors, Jan. 2026
 E. Turan, G. Abel, M. Behmanesh, E. Pierson, M. Ovsjanikov.
 Published in ICML 2026 (spotlight: 2% of top submissions).

Reducing Recurrent Competitive Epidemics via Dynamic Resource Allocation Dec. 2025
 A. Kalogeratos, G. Abel, S. Sarao Mannelli
 Submitted in *IEEE Transactions on Network Science and Engineering*

MultiLayerNetViz : A Multi-Layered Visualization Module for Information Nov. 2025
Diffusion in Social Networks. G. Abel
 Python module for multi-layered networked data (available on Github)

SELECTED TALKS

frCSS 2025 French Local Chapter - Conference on Complex Systems/Sunbelt Apr. 2024
CSS 2025 International Conference on Complex Systems, Siena Sept. 2025
ARCOM International day for Social Media Research Nov. 2025
 (French Authority for the Regulation of Communication in Media)
Conference for Young Applied Maths scientists. Mar. 2026
 Champs-sur-Marne, France

TEACHING EXPERIENCE

Descriptive and Inferential Statistics
 1st year of Cycle Pluridisciplinaire en Enseignement Supérieur (CPES), 2024-2026
 Paris-Saclay University. Creation of the syllabus, teaching courses,
 exercise sessions and exams.
 Total: 192 hours of teaching: 117h of course + 75h of exercise sessions.

ACADEMIC EXPERIENCE

Internship Supervision:
 Mahdi HADJ-TAIEB, M1 ENSTA student June - Aug. 2025
Sequential learning for Temporal Point Process models
 Mohamed BENLOUGHMARI, M1 ENSTA student Oct. - Dec. 2025
Density-based curriculum learning for Temporal Point Processes

Official reviewing:
 Journal article. *IEEE Transactions On Network Science and Engineering,* June 2026
 Journal article. *IEEE Transactions On Network Science and Engineering,* June 2026
Sub-reviewing:
 Journal article. *IEEE Transactions On Computation Social Systems,* Oct. 2025
 Conference paper. *International Conference on Machine Learning (ICML)* Feb. 2026

AWARDS & GRANTS

PhD Grant for students from the ENS Paris-Saclay Sep. 2023
Winner of the Hackathon MathTech FMJH Dec. 2023
 1 week challenge organised by EDF on forecasting energy consumption